Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Mark Courage	Email	Mark.courage@tno.nl
Institute Name	TNO (Netherlands Organisation for Applied Scientific Research)			
Address	Stieltjesweg 1, 2628 CK Delft, The Netherlands			

2. Technology Overview

Title of Proposed Technology	Hollow-Core Fiber Photothermal Spectroscopic Gas Sensor for Hydrogen Fuel Quality Control			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical・Electronics		Energy & Resources	
	Medium Category	Small Category	Medium Category	Small Category
	Measuring Instruments	Optical Measuring Instruments	New & Renewable Energy	Hydrogen
Concept of Proposed Technology	Proton-exchange-membrane (PEM) fuel cells are acutely sensitive to trace contaminants (ppb–ppm), which can irreversibly poison catalysts (e.g., CO, sulfur species) or degrade membranes (e.g., NH_3, halides). As a result, hydrogen fuel for mobility must meet stringent specifications codified in **ISO 14687** [1] and the harmonized **SAE J2719** standards [2], including sub-ppm thresholds for CO (≤ 0.2 μmol mol^{-1}), total sulfur (≤ 0.004 μmol mol^{-1}), and water (≤ 5 μmol mol^{-1}), among others. These specifications are complemented by guidance on analytical methods (ISO 21087) and sampling (ISO 19880-8), and are now enforced across refueling stations with accredited laboratories (e.g., NPL) providing traceable measurements and reference mixtures. Purity control in real deployments is not achieved with a single instrument; rather, stations and producers combine **on-line sensors** for trend monitoring with **reference-grade analyzers** for certification. TNO's hollow core fiber photothermal spectroscopy sensing technology enables real-time, multi-gas analysis as an on-line sensor.			

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Photothermal spectroscopy (PTS) detects **temperature-induced refractive-index/phase changes** in a gas caused by **absorption of a pump beam**; a probe beam (often near-IR) interrogates the induced phase shift. When implemented in **hollow-core fibers**, PTS achieves **long, confined light–gas interaction** with small sample volumes, enabling **high sensitivity** while avoiding direct contact of optics with the gas stream.

This significantly decreases the footprint of the gas sensor which the HCF as gas sensor with a small dimension (sub mm diameter and ~ 100 mm in length) can be flexibly installed in difficult accessible location with extreme environmental conditions e.g. temperature, pressure, toxic gas, explosion danger. The HCF gas sensor is based on the same photothermal effect as PAS. In contrast to PAS, the pump laser induced change of refractive index of the molecule to be detected is used [3].

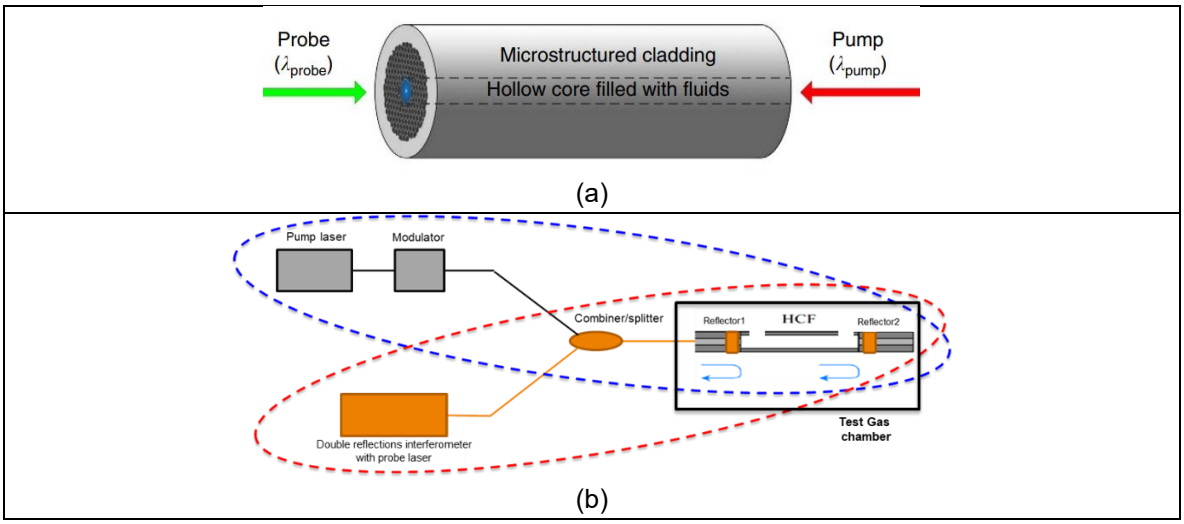

Fig. 1 Hollow core fiber for molecule in fluids medium detection and pump laser light interaction to generate photothermal effect [2] (a). The proposed HCF concept with the pump laser with modulated amplitude the generate the refractive index change in the HCF (blue dashed area), the connecting fiber to the HCF with inlets for gas, and the fiber interferometer detection module with the probe laser (red dashed area) [3-5] (b).

The proposed HCF gas sensor consists of 3 main parts [3] (see Fig. 1(b)):

- The pump laser module to generate the refractive index change in the HCF
- The HCF with inlet for gas from the environment. The HCF will be connected to stand commercial off the self (COTS) telecom fiber including two reflectors.
- The interferometer detection module to measure the change in the HCF optical path length using the reflections from the two reflectors.

Optical interferometer [4] is one of the most sensitive technologies to measure path length changes and is used in the proposed HCF gas sensing system.

Since laser absorption technology is used to measure the absorption of the gas, changing the wavelength of the pump laser only in the HCF gas sensor configuration in Fig. 1b will enable the measurement of gases with different absorption spectrum. Furthermore, using standard COTS telecom components, a network of multiple HCF sensors can be realized sharing the same pump laser module and interferometer detection module to extent the coverage of the gas sensing system.

Rationale for Joint R&D Project

The global hydrogen economy is transitioning from pilot infrastructure to commercial-scale deployment, particularly in mobility, electrolyzers, and import/export corridors. While hydrogen production, compression, and storage technologies are advancing rapidly, hydrogen fuel quality assurance remains a systemic bottleneck:

- Fuel-cell systems demand ppb-level impurity control (CO, sulfur species, NH_3, H_2O)

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	• Current compliance relies heavily on offline laboratory analysis • Continuous, multi-species, in-situ sensing solutions remain technically immature and commercially unavailable As hydrogen trade becomes international (e.g. liquefied hydrogen, ammonia cracking, pipeline blending), fuel quality risk shifts from local to transnational, creating an urgent need for robust, real-time, certifiable sensor technologies. Hollow-core-fiber photothermal spectroscopy (HCF-PTS) is one of the few emerging technologies with the theoretical capability to bridge this gap—but it is not yet industrially deployable. Addressing this requires coordinated expertise across metrology, photonics, system engineering, and manufacturing that no single organization or country currently holds. TNO holds key IPs on HCF-PTS system including: patents [3-5], dedicated tools and engineering methods [6], most advanced research and testing labs and facilities and more than 40 years Know-Hows in Fiber Optics domain. South Korea has one of the world's highest concentrations of fuel-cell vehicles, large-scale hydrogen refuelling station roll-out and strong government push toward hydrogen import chains. This creates a real pull from end users, ensuring that sensor development is guided by practical operational needs, not only research metrics.
Key Words	1. Gas sensing and analysis 2. Fiber optic sensor 3. Photoacoustic spectroscopy 4. Fiber interferometer 5. Hydrogen fuel quality

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)

R&D Objectives

- Develop an HCF sensor architecture optimised for hydrogen matrices
- Demonstrate ppb–ppm sensitivity for selected critical impurities (e.g. CO, NH_3, H_2O)
- Validate sensor performance under realistic hydrogen infrastructure conditions
- Prepare a manufacturable sensor module concept with a credible commercial pathway

R&D Target Specifications

Optical interferometer [1] is one of the key technologies for high performance position/displacement sensor and provides (the combination of) high resolution, high-speed, large dynamic range non-contact measurement. Different conventional interferometer configurations (including Michelson, Mach-Zehnder, Fabry-Perot, white light interferometer [5]) are known.

Current status of the proposed HCF gas sensor is a proof-of-concept demonstrator setup realized for CO2 detection.

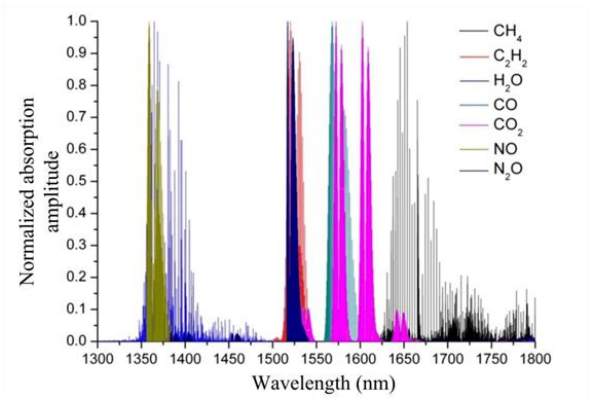

Fig. 2 Normalized absorption spectrum of different gases around 1550 nm.

Among the different gases with absorption around the telecom 1550 nm [xx] , CO2 is selected for the inherent safety for the lab. The absorption of CO2 about 1575 nm is used because of the availability of standard telecom wavelength components. Despite the CO2 absorption at this wavelength is very small, the detection of the CO2 absorption spectrum between 1565 nm and 1575 nm was demonstrated and good with the Hitrand spectrum was found. By locking the wavelength to one of the absorption wavelengths, the sensitivity of the system can be calibrated and used for gas concentration measurement.

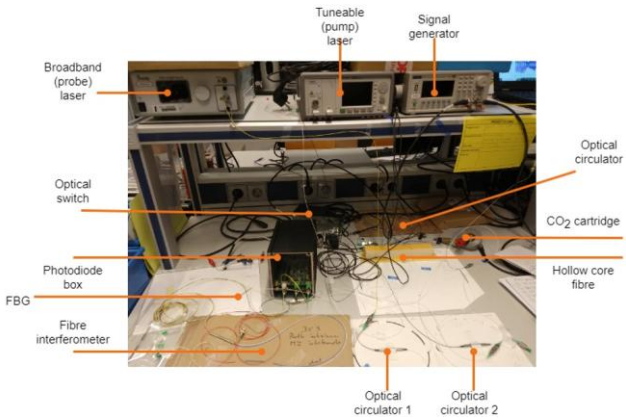

Fig. 3 Lab test setup of the HCF for CO2 detection demonstration.

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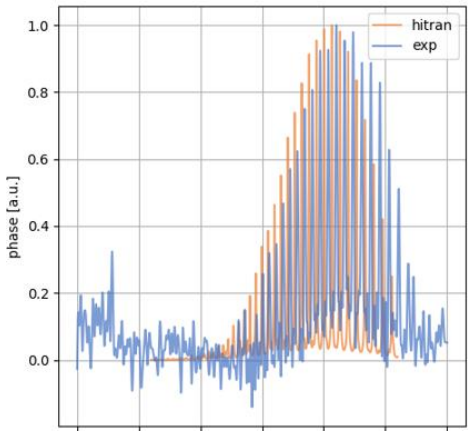

Fig. 4 Test results of CO2 absorption spectrum measurement between 1565 and 1575 nm and comparison with Hitran (TNO internal report).

R&D Plan

For the technology development from TRL-3 to TRL-5 the following developments are necessary, and a TRL-relevant work plan is made.

WP1 — Requirements, Standards Mapping & Use-Case Definition
Lead: TNO | **Support:** Korean SMEs
Months: 1–4
Key Tasks

1. Requirements capture**:**
2. Use-case(s) defined by SME(s)
3. Risk & compliance pre-assessment

Deliverables

- **D1.1** Requirements & Standards Matrix (ISO 14687 / EN 17124 → sensor targets)
- **D1.2** Use-case & test-environment definition report.

WP2 — Concept & Spectral Feasibility (TRL-3)
Lead: TNO | **Support:** Korean SMEs
Months: 3–8
Key Tasks

1. **Spectral line survey** for target impurities (e.g., H_2O ~1.39 µm, CO ~2.3–2.35 µm, CH_4 ~3.3 µm, CO_2 ~2.0–2.7 µm, H_2S/NH_3 in MIR), with pump availability and selectivity checks.
2. **Photothermal modeling** inside HCF core (thermal conduction, relaxation dynamics, phase modulation) to estimate NEC/LOD and response time. (Modeling guided by PTI in HCF literature.)
3. **Architecture down-select**: HCF type (anti-resonant/negative-curvature vs. PBG), interferometer (FPI vs. DMI), multiplexing strategy (FDM for pumps).

Deliverables

- **D2.1** Feasibility report with modeled performance (NEC/LOD, dynamic range).
- **D2.2** Preliminary optical/mechanical design pack (pump–probe lines, HCF selection, readout).

WP3 — Single-Species Proof-of-Concept (TRL-3)
Lead: TNO
Months: 7–12
Key Tasks

1. Build lab test demo: short HCF cell, single pump line, telecom-band probe, lock-in detection.
2. Demonstrate repeatable photothermal signal vs. concentration curve in hydrogen background for at least one species (e.g., H_2O or CO).
3. Perform noise & stability analysis (Allan variance, drift sources) and validate model.

Deliverables

- **D3.1** PoC dataset & report (repeatability, LOD estimates, uncertainties).

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- **D3.2** Updated design requirements (optical power, modulation parameters).

WP4 — Integrated Lab Prototype (Multi-Impurity) (TRL-4)
Lead: TNO | **Support:** Korean SMEs
Months: 10–20
Key Tasks

1. System integration**:** Compact interferometric readout (HCF-FPI), stable cavity locking (OPMA if feasible), multi-pump multiplexing (≥ 3 species).
2. Calibration & method validation**:** Controlled gas mixtures; linearity, repeatability, cross-sensitivity mapping; uncertainty budgets aligned with QA principles in EN 17124 (non-certifying at TRL-4).
3. Environmental tests**:** Temperature and pressure variations in lab; hydrogen matrix effects; mechanical robustness (vibration/shock).

Deliverables

- **D4.1** TRL-4 prototype (bench-top) with multi-impurity measurement capability.
- **D4.2** Calibration curves & uncertainty report; cross-sensitivity matrix.
- **D4.3** Engineering risk register & mitigation plan.

WP5 — Relevant-Environment Breadboard Validation (TRL-5)
Lead: Korean SMEs (test stand owner) | **Support:** TNO
Months: 18–28
Key Tasks

1. Assemble breadboard with ruggedized optics/electronics (not final enclosure); SCADA-style data logging; automated sequences.
2. Validate performance in simulated refueling conditions: hydrogen background, elevated pressure (e.g., 10–30 bar), realistic flow rates, temperature cycling—no live dispenser at this stage.
3. Long-term stability (days–weeks), response/recovery times, impurity step changes; compare to lab references (metrology partner) to quantify accuracy.
4. Standards readiness review: Gap analysis to ISO 14687 targets and EN 17124 QA concepts for a future TRL-6 pilot.

Deliverables

- **D5.1** TRL-5 validation report (performance vs. targets).
- **D5.2** Reliability & maintainability assessment (calibration intervals, drift compensation).
- **D5.3** TRL-6 pilot plan (requirements, risks, safety interfaces).

WP6 — Project Management, IP & Exploitation
Lead: TNO | **Support:** All
Months: 1–30 (continuous)
Key Tasks

- PMO, progress reviews, risk & quality assurance; IP strategy (foreground/background); data management.
- Exploitation roadmap: pre-certification steps toward ISO/EN-aligned validation at pilot stations; Korean and EU market pathways, leveraging national interest in real-time impurity monitoring**.**

Deliverables

- **D6.1** Quarterly progress reports & risk registers.
- **D6.2** IP & Exploitation Plan (TRL-6 onward).
- **D6.3** Final technical report.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TNO will act as the system-level research lead and technology integrator, leveraging its background IP in HCF-PTS gas sensing and its expertise in hydrogen systems and metrology. TNO as the independent R&D research institute is responsible for • Defining and maturing the optical fiber sensing concept into a deployable monitoring/diagnostic subsystem for hydrogen purity control at selected use case and scenario. • Bridging fundamental sensing science and engineering knowledge, reducing technical risk for SMEs by delivering validated sensing methods, integration guidelines, and diagnostic performance evidence. • Ensuring sensor development, and validation and implementation using test protocols agreed with Korean partners. TNO holds key IPs on HCF-PTS system including: 2 patents, dedicated tools and engineering methods, most advanced research and testing labs and facilities and more than 40 years Know-Hows in Fiber Optics domain.
Expected R&R of Korean SME Partner	The South Korean SME partner(s) will act as the product development, industrialisation, and commercialisation lead(s) within the consortium. SME1: Sensor product development company or hydrogen fuel quality monitoring and control engineering company SME2: Fuel cell OEM or hydrogen fuel supplier (e.g. refueling station operation) – realistic testing facility and provide user specification and requirement SMEs are expected to bring knowledge and work on: • Translate the HCF sensing concepts and system requirements defined by TNO into engineered, manufacturable sensor components and modules • Drive the project from laboratory feasibility toward deployable prototypes • Operation profiles, use cases, and market requirements • Field trials, application validation, and productization pathways • System controller integration and commercialization

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	Technological • TRL-5 validated HCF photothermal gas sensor system • Multi-impurity hydrogen quality sensing demonstrated • Standards-aligned measurement framework and TRL-6 roadmap Commercial • Clear commercialization pathway for Korean SMEs • Differentiated sensor technology for hydrogen infrastructure • Reduced risk for pilot deployment and scale-up Industrial & Economic • Safer and more reliable hydrogen refueling infrastructure • Strengthened EU–Korea innovation ecosystem • Contribution to hydrogen economy growth and decarbonization goals

6. References

1. ISO, *ISO 14687:2025 Hydrogen fuel quality — Product specification*. 2025.
2. SAE, *SAE J2719 Surface Vehicle Standard*. 2020.
3. Cheng, L.-K. and M. Xu, *A Measurement System and Method for Detecting A Molecule Concentration in A Medium*, Patent WO2024186209(A1) 2024.
4. Cheng, L.-K. and M. Xu, *Optical Sensor System For Detection Of A Chemical Substance*, patent EP4650723A1; WO2025239776A1 2025.
5. Cheng, L.-K., M. Xu, and M. Eschen, *An Interrogator System and A Method*, patent EP4679040A1; WO2026015025A1, 2026.
6. Cheng, L.K., et al. *Fiber interferometer combining sub-nm displacement resolution with miniaturized sensor head*. in *2017 25th Optical Fiber Sensors Conference (OFS)*. 2017.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization